

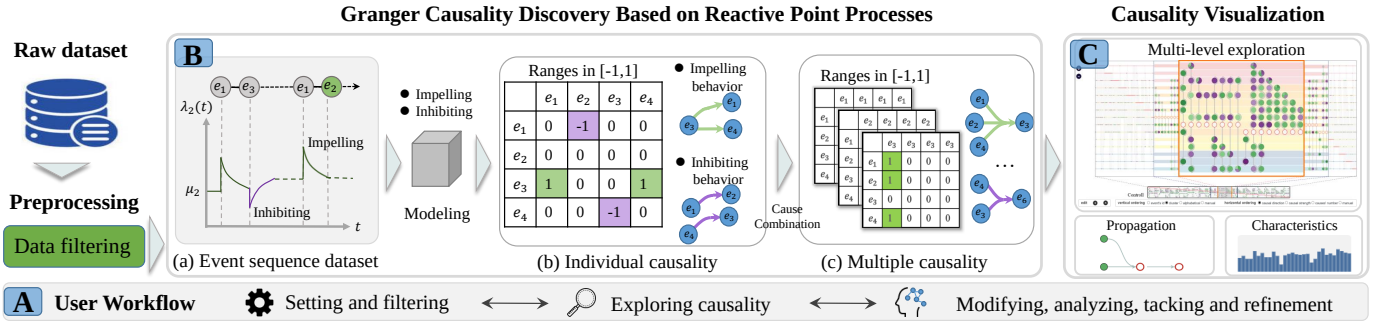


Fig. 2. The system architecture (A) and the pipeline of our combined causal discovery algorithm (B) which impelling and inhibiting reveals behaviors. The impelling and inhibiting behaviors discovery is based on granger causality, which consists of three key steps: (a) training a reactive point processes model to fit event sequence dataset, (b) discovering impelling or inhibiting behaviors on causality, which ranges in $[-1, 1]$ between two entity, and (c) inferring the cause combinations based on the individual causal discovery. The framework of our individual and combined causality analysis system (C) can be divided into the following steps: preprocessing and filtering raw dataset, modeling combined causal discovery and exploring directed hypergraph with diverse ordering operations and interaction techniques.

ing a complex causality is time-consuming, thus, diverse causal patterns must be illustrated and highlighted to help users significantly improve exploration efficiency. [50] [20] CausalFlow [51]

T6 Exploring causal relations at a multilevel. Users may need to involve different levels of concepts, i.e., the overall trend of causality and focused causality, which is a complex scenario. Hence, a multi-level exploration can help users organize their analysis and improve exploration efficiency.

These requirements help us obtain appropriate visual design principles and make informed decisions about our visual design.

3.2 Model Pipeline and User Workflow

We proposed a combined causal discovery algorithm for detecting impelling and inhibiting behaviors shown in Figure 2(B). After extracting useful entities and filtering valid event sequences from the raw dataset, Reactive point processes is employed to model Granger causality for inferring the impelling or inhibiting impact among entities in temporal event sequences. In this process, eliminating and recruiting principles on candidate cause combinations are defined for ensuring the effectiveness and controllability on cause combinations. Based on above operations, we obtain a weighted and directed hypergraph dataset, which uncovers the impelling and inhibiting behaviors among entities. To visually analyze causality, We design an interactive causality analysis system shown in Figure 2(C) for supporting the user analysis task (T1-T6) with the system architecture and user workflow featured in Figure 2(A). The system shown in Figure 1 includes three panels, namely, combined causality panel, propagation panel, and parameters configuration panel. With this system, a user can start with an overview of a novel directed parallel-based causality visualization. Then, causal patterns are presented in directed and parallel-based visualization with various ordering operations and aggregation strategy for highlighting significant impelling and inhibiting impact patterns. Flexible interactions, such as filtering causal strength, brushing, and selecting focused causality, are provided to explore causal relations at different levels of causality abstraction. Next, users are supported to further modify the incorrect causal relations, analyze impelling and inhibiting behavior patterns, refine and track the causal influence and propagations aided by various ordering operations, smooth interactions.

4 CAUSALITY MODEL

We employ Reactive point processes to model Granger causality in event sequences. Then, the causality model are trained to fit event sequence dataset by optimizing objective function. We utilize parameters of the trained model to infer causal strength, as well as impelling or inhibiting behaviors of causality between events. Finally, cause combination rules are defined to detect combined causes with one-to-one-relationship causal discovery algorithm.

4.1 Granger Causality on Reactive Point Processes

Inspired by causal discovery [20], which only uncovers impelling impact among events, this section describes the extended model for revealing impelling and inhibiting impact among events. The causal discovery [20] use Hawkes processes to establish a model by relating impelling events. The occurrence probability of event u is inferred from its conditional intensity function:

$$\lambda_u(t) = \mu_u + \sum_{u'=1}^U \int_0^t \phi_{uu'}(s) dN_{u'}(t-s) \quad (1)$$

where the first constant term μ_u denotes basic entity intensity and the second term represents the influence of past event on entity u at time t . The value of inferring impact factor is positive, which is a parameter of obtained from the trained Hawkes processes model. Reactive point processes [13] [57] is an extension of Hawkes processes by relating two types of impelling and inhibiting events. Hawkes processes only considers the impelling impact of events, while Reactive processes model both impelling and inhibiting effects of chronological events on instantaneous intensity, which can be written as follows:

$$\lambda_u(t) = \mu_u + \sum_{u'=1}^U \int_0^t \phi_{uu'}(s) dN_{u'}(t-s) - \sum_{u'=1}^U \int_0^t \psi_{uu'}(s) dN_{u'}(t-s) \quad (2)$$

where $\phi_{uu'}$ and $\psi_{uu'}$ are kernel functions to reflect the impelling or inhibiting impacts from event u to u' respectively. The intensity function $\lambda_u(t)$ combines an inhibiting term for accounting inhibiting effects from past entities and describing the causality intensity among entities. Given that $\lambda_u(t)$ describes the causal influence intensity among entities, the value of $\lambda_u(t)$ must be positive to ensure the intensity nonnegativity of events occurrence. Many researchers utilize a nonlinear function [54] [57] to guarantee negativity. Similarly, we leverage a nonlinear function $\hat{\lambda}_u(t)$ to guarantee the negativity of $\lambda_u(t)$.

$$\hat{\lambda}_u(t) = s \log \left(1 + e^{\frac{\lambda_u(t)}{s}} \right) \quad (3)$$

where s is a small positive constant number, which ensures that we can utilize the property function $g(x) = s \log(1 + \exp(x/s)) \approx \max\{0, x\}$.

Then, we handle impact function as a linear combination of basis kernel functions, written as follows:

$$\begin{aligned} \phi_{uu'}(t) &= \sum_{m=1}^M a_{uu'}^m \kappa_m(t) \\ \psi_{uu'}(t) &= \sum_{m=1}^M b_{uu'}^m \kappa_m(t) \end{aligned} \quad (4)$$